

Lab Assignment 11: AI Concepts & Architectures

Course: PAI Lab | Task ID: 11

I. Foundational Models and Architectures

1. Generative AI (GenAI)

Generative AI is a broad category of artificial intelligence focused on creating new data. Unlike traditional AI that classifies or predicts based on existing data, GenAI uses patterns learned during training to produce original content such as text, images, code, and audio.

2. LLMs (Large Language Models)

LLMs are advanced AI models specifically engineered to process and generate human language. Built on transformer architectures and trained on vast datasets, they excel at understanding context and engaging in coherent dialogue. Examples include GPT-4 and Llama 3.

3. GANs (Generative Adversarial Networks)

GANs consist of two competing neural networks: the generator and the discriminator. The generator creates synthetic data, while the discriminator evaluates its authenticity. This process continues until the generator produces output indistinguishable from real data.

II. Application Frameworks and Retrieval

4. LangChain

LangChain is a specialized framework designed to simplify the creation of applications powered by large language models. It provides the orchestration logic needed to link LLMs with external components like memory, APIs, and document loaders.

5. RAG (Retrieval-Augmented Generation)

RAG is an architectural approach that enhances LLM responses by fetching relevant information from a specific knowledge base before generating an answer. This allows the model to provide accurate details from private documents without requiring retraining.

III. Data Representation and Search

6. Vector

In AI, a vector is a mathematical representation of data in a multi-dimensional space. These numerical arrays capture semantic meaning; for instance, vectors for conceptually related words will be mathematically closer to each other.

7. VectorDB (Vector Database)

A Vector Database is a storage system optimized for handling high-dimensional vectors. Unlike traditional databases that search for exact matches, a VectorDB allows for similarity searches, finding data points that are conceptually related to a query.

8. FAISS (Facebook AI Similarity Search)

FAISS is a library built for efficient similarity search and clustering of dense vectors. It provides high-performance algorithms that can search through massive datasets in milliseconds, often serving as the engine for vector search systems.

Concept	Primary Function	Simple Analogy
GenAI	Content Creation	The Art Gallery
LLM	Language Reasoning	The Professional Writer
LangChain	Workflow Orchestration	The Project Manager
RAG	Information Grounding	An Open-Book Exam
VectorDB	Semantic Storage	A Context-Aware Filing Cabinet